Software Requirements Specification

Version 1.0

April, 20, 2026

KiwiGo

Saria Kabbour

Table of Contents

1.0. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the KiwiGo ride-share application. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate, and how the system will react to external stimuli. This document is intended for both the stakeholders and the developers of the system, including project managers, developers, testers, and end users.

1.2. Scope of Project

KiwiGo is a mobile ride-share app that connects customers with nearby drivers. Customers can request rides, track drivers in real time, and make payments through the app. Drivers can view and accept ride requests, navigate to pickup locations, and manage their availability. The system is designed to provide a safe, reliable, and convenient transportation service.

1.3. Overview of Document

The next section, the Overall Description, provides an overview of the KiwiGo system functionality through a use case diagram and a short description of the system's actors and interactions. Section 3 contains the detailed requirements specification, which includes the functional requirements with detailed use case descriptions, the logical database design with an ER diagram, and the non-functional requirements covering performance, security, and availability.

2.0. Overall Description

2.1 Use case summary

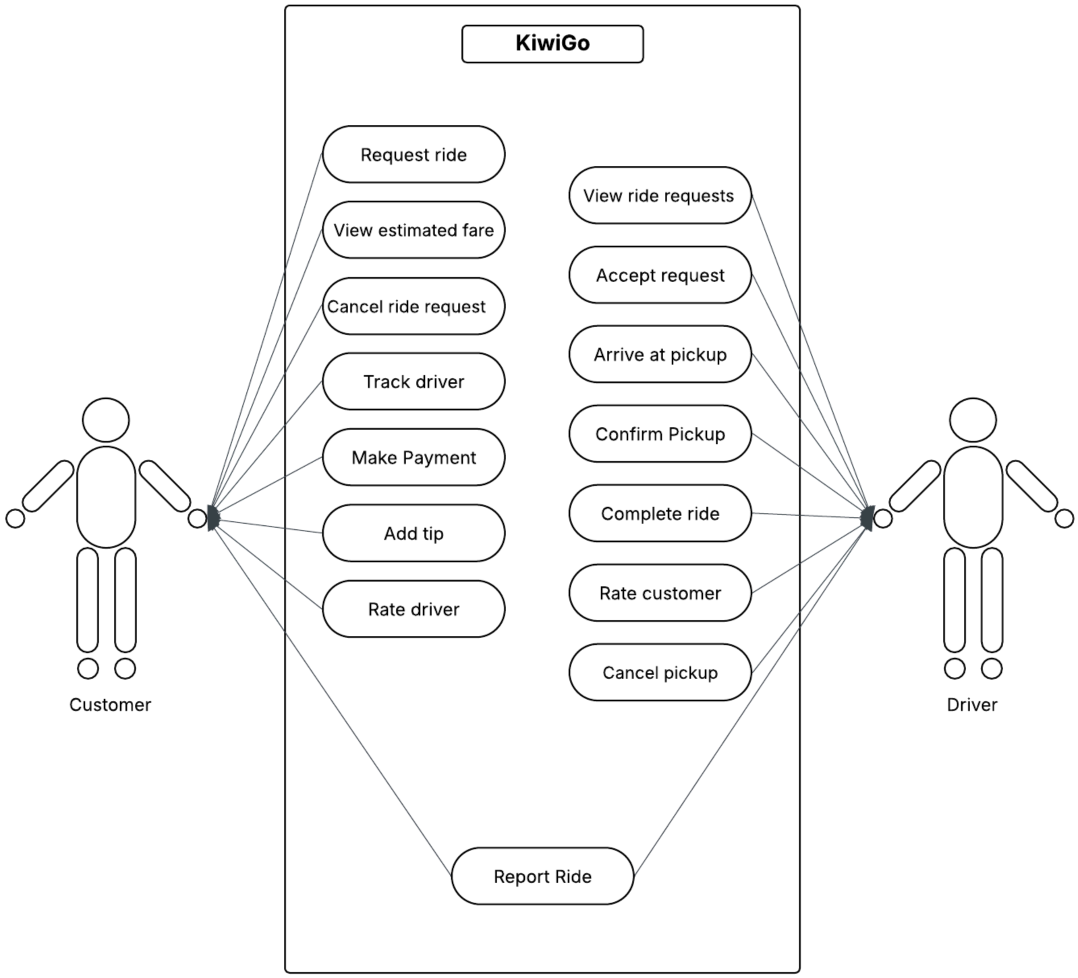

Figure 1 – Use case summary

The KiwiGo ride-share application has two active actors: the Customer and the Driver. The Customer accesses the system through a mobile app to request rides, track drivers, and make payments. The Driver accesses the system through a separate mobile app interface to view and accept ride requests, manage pickups, and complete rides. Both actors can report unsatisfactory ride experiences through the app.

2.2 Functional Requirements Specification

This section outlines the use cases for each actor separately. The Customer initiates the ride process while the Driver responds to and fulfills ride requests.

The KiwiGo Ride Process consists of the use cases listed below:

- A Customer opens the app and requests a ride, specifying their destination.
- The system calculates and displays the estimated fare to the Customer.
- A Driver views available ride requests and accepts one.
- The system notifies the Customer that a Driver is on the way.
- The Driver arrives at the pickup location and confirms the pickup.
- The Driver completes the ride and drops off the Customer at the destination.
- The Customer makes a payment and may add a tip through the app.
- Both the Customer and Driver may rate each other and report rides.

2.2.1 Customer Use Cases

Ride Management consists of use cases:

- Request ride
- View estimated fare
- Cancel ride request
- Track driver

Payment and Feedback consists of use cases:

- Make payment
- Add tip
- Rate driver
- Report ride

2.2.2 Driver Use Cases

Ride Fulfillment consists of use cases:

- View ride requests
- Accept request
- Arrive at pickup
- Confirm pickup

Ride Completion consists of use cases:

- Complete ride
- Rate customer
- Cancel pickup
- Report ride

2.3 User Characteristics

The Customer is expected to be mobile app literate and comfortable using GPS-based applications on a smartphone. The Customer should be able to navigate a simple interface to request rides, view fare estimates, and make digital payments through the app.

The Driver is expected to be mobile app literate and comfortable using navigation and GPS tools on a smartphone. The Driver should be able to manage ride requests, navigate to pickup and drop-off locations, and interact with the app while following safe driving practices.

3.0. Requirements Specification

3.1 External Interface Requirements

The KiwiGo system interfaces with two external systems. First, a GPS/Maps API (such as Google Maps) is used to determine the current location of the customer and driver, calculate estimated travel times, and provide navigation to the driver. Second, a

Payment Processing API (such as Stripe) is used to handle all financial transactions including ride payments and tips. The system sends the ride fare and payment details to the payment processor and receives a confirmation of the transaction in return.

3.2 Functional Requirements

The Logical Structure of the Data is described in Section 3.3. The Xref entry cross references the requirements to a use case from Section 2.

Field	Description
Use Case Name	Request Ride
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Customer opens the KiwiGo app and requests a ride.
Precondition	The Customer is logged into the app and GPS location is enabled.
Basic Path	1. The Customer opens the KiwiGo app. 2. The system uses GPS to detect the Customer's current location. 3. The Customer enters their desired destination. 4. The system calculates and displays the estimated fare and arrival time. 5. The Customer confirms the ride request. 6. The system searches for available Drivers in the area. 7. The system notifies the Customer that a Driver has been found.
Alternative Paths	In step 6, if no Drivers are available in the area, the system notifies the Customer and the request is cancelled.

Field	Description
Postcondition	A ride request has been submitted and a Driver has been matched to the Customer.
Exception Paths	The Customer may cancel the request at any time before a Driver is matched.
Other	The fare estimate is based on distance, time of day, and current demand.

3.2.2 View Estimated Fare

Field	Description
Use Case Name	View Estimated Fare
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Customer enters a destination in the KiwiGo app
Precondition	The Customer is logged into the app and has entered a pickup and destination location
Basic Path	1. The Customer enters their desired destination. 2. The system calculates the estimated fare based on distance and time of day. 3. The system displays the estimated fare to the Customer. 4. The Customer reviews the fare and chooses to confirm or cancel the ride request.

Alternative Paths	In step 2, if the fare cannot be calculated due to an invalid destination, the system displays an error message and prompts the Customer to re-enter a valid destination.
Postcondition	The Customer has viewed the estimated fare and made a decision to confirm or cancel the ride request.
Exception Paths	The Customer may abandon the fare view and exit the app at any time.
Other	The fare may vary based on peak hours and high demand periods. The final charge may differ slightly from the estimate based on actual trip duration and distance.

3.2.3 Cancel Ride Request

Field	Description
Use Case Name	Cancel Ride Request
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Customer selects the cancel option in the KiwiGo app after submitting a ride request
Precondition	The Customer has an active ride request and the Driver has not yet arrived at the pickup location

Basic Path	1. The Customer selects the cancel option in the app. 2. The system prompts the Customer to confirm the cancellation. 3. The Customer confirms the cancellation. 4. The system cancels the ride request. 5. The system notifies the Driver that the ride has been cancelled. 6. The system updates the ride status to cancelled in the database.
Alternative Paths	In step 3, if the Customer does not confirm the cancellation, the system returns to the active ride screen and the ride continues as normal.
Postcondition	The ride request has been cancelled and the Driver has been notified. The Driver is now available to accept other ride requests.
Exception Paths	The Customer may abandon the cancellation process at any time and return to the active ride screen.
Other	The system monitors Customers who frequently cancel ride requests. Repeated cancellations may result in a warning or account suspension.

3.2.4 Track Driver

Field	Description
Use Case Name	Track Driver

Xref	Section 2.2.1 Customer Use Cases
Trigger	A Driver has accepted the Customer's ride request
Precondition	The Customer has an active ride request and a Driver has been matched and is on the way
Basic Path	1. The system notifies the Customer that a Driver has accepted the request. 2. The system displays the Driver's current location on a map in real time. 3. The system displays the Driver's estimated time of arrival. 4. The Customer monitors the Driver's location as the Driver travels to the pickup location. 5. The system updates the Driver's location on the map continuously until the Driver arrives. 6. The system notifies the Customer when the Driver has arrived at the pickup location.
Alternative Paths	In step 3, if the Driver is taking significantly longer than the estimated arrival time, the system updates the estimated arrival time and notifies the Customer.
Postcondition	The Driver has arrived at the pickup location and the Customer has been notified.

Exception Paths	The Customer may cancel the ride at any time while tracking the Driver before the Driver arrives.
Other	The tracking feature requires GPS to be enabled on both the Customer's and Driver's mobile devices.

3.2.5 Make Payment

Field	Description
Use Case Name	Make Payment
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Driver marks the ride as complete in the KiwiGo app
Precondition	The ride has been completed and the Customer has been dropped off at the destination
Basic Path	1. The system calculates the final fare based on the actual distance and duration of the ride. 2. The system displays the final fare to the Customer.

	3. The system automatically charges the Customer's saved payment method. 4. The system sends a payment confirmation to the Customer. 5. The system sends a payment notification to the Driver. 6. The system updates the ride status to completed and paid in the database.
Alternative Paths	In step 3, if the payment fails, the system notifies the Customer and prompts them to update their payment method or enter a new one.
Postcondition	The payment has been processed and both the Customer and Driver have been notified of the transaction.
Exception Paths	If the payment cannot be processed after multiple attempts, the system flags the account and notifies the Customer to resolve the issue.
Other	All payment transactions are processed through an external payment processing API. All financial data is encrypted and securely transmitted.

3.2.6 Add Tip

Field	Description
Use Case Name	Add Tip
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Customer selects the option to add a tip after the ride has been completed
Precondition	The ride has been completed and the Customer has received a payment confirmation
Basic Path	1. The system displays a tip prompt to the Customer after the ride is completed. 2. The system presents the Customer with suggested tip amounts as a percentage of the fare. 3. The Customer selects a tip amount or enters a custom tip amount. 4. The Customer confirms the tip. 5. The system processes the tip through the payment API. 6. The system sends a tip confirmation notification to the Customer and Driver. 7. The system updates the total payment in the database.

Alternative Paths	In step 3, if the Customer does not wish to leave a tip, the Customer may select no tip and proceed to rate the Driver.
Postcondition	The tip has been processed and added to the Driver's total earnings for the ride.
Exception Paths	The Customer may skip the tip prompt at any time and proceed to the ride rating screen.
Other	Tips are optional and are processed separately from the base fare. The Driver receives 100% of the tip amount.

3.2.7 Rate Driver

Field	Description
Use Case Name	Rate Driver
Xref	Section 2.2.1 Customer Use Cases
Trigger	The Customer is prompted to rate the Driver after the ride has been completed

Precondition	The ride has been completed and the payment has been processed
Basic Path	1. The system displays a rating prompt to the Customer after the ride is completed. 2. The Customer selects a rating between 1 and 5 stars for the Driver. 3. The Customer may optionally enter a written comment about the ride experience. 4. The Customer submits the rating. 5. The system saves the rating and updates the Driver's overall rating in the database. 6. The system confirms to the Customer that the rating has been submitted.
Alternative Paths	In step 2, if the Customer selects a rating of 1 or 2 stars, the system prompts the Customer to provide additional details about the unsatisfactory experience.
Postcondition	The Driver's rating has been updated in the system and the ride record is marked as fully complete.
Exception Paths	The Customer may skip the rating prompt at any time. The ride will still be marked as complete in the system.

Other	The system will not pair a Customer and Driver together on future rides if either party has previously rated the other as unsatisfactory.

3.2.8 Report Ride

Field	Description
Use Case Name	Report Ride
Xref	Section 2.2.1 Customer Use Cases; Section 2.2.2 Driver Use Cases
Trigger	The Customer or Driver selects the option to report an unsatisfactory ride experience
Precondition	The ride has been completed and the Customer or Driver is logged into the KiwiGo app
Basic Path	1. The Customer or Driver selects the report option in the app. 2. The system displays a list of reasons for the report such as unsafe driving, inappropriate behavior, or no show.

	3. The Customer or Driver selects the appropriate reason for the report. 4. The Customer or Driver may optionally enter additional details about the incident. 5. The Customer or Driver submits the report. 6. The system records the report in the database and assigns it a case number. 7. The system sends a confirmation notification to the reporting party.
Alternative Paths	In step 3, if the reason for the report is not listed, the Customer or Driver may select other and provide a written description of the incident.
Postcondition	The report has been recorded in the system and will be reviewed by KiwiGo support staff.
Exception Paths	The Customer or Driver may abandon the report at any time without submitting.
Other	The system monitors the number of reports filed against each Customer and Driver. Accounts with a high number of reports may be suspended or permanently banned from the platform.

3.2.9 View Ride Requests

Field	Description
Use Case Name	View Ride Requests
Xref	Section 2.2.2 Driver Use Cases
Trigger	The Driver opens the KiwiGo driver app and sets their status to available
Precondition	The Driver is logged into the app and GPS location is enabled on their device
Basic Path	1. The Driver opens the KiwiGo driver app and sets their status to available. 2. The system detects the Driver's current location using GPS. 3. The system displays a list of available ride requests within a certain radius of the Driver's current location. 4. For each ride request, the system displays the Customer's pickup location, destination, and estimated fare. 5. The Driver reviews the available ride requests and selects one to accept.

Alternative Paths	In step 3, if there are no available ride requests in the Driver's area, the system displays a message indicating no rides are currently available and continues to monitor for new requests.
Postcondition	The Driver has viewed the available ride requests and is ready to accept one.
Exception Paths	The Driver may set their status to unavailable at any time to stop receiving ride requests.
Other	The system will not display ride requests from Customers who have previously rated the Driver as unsatisfactory, and vice versa.

3.2.10 Accept Request

Field	Description
Use Case Name	Accept Request
Xref	Section 2.2.2 Driver Use Cases

Trigger	The Driver selects a ride request from the list of available requests
Precondition	The Driver is logged into the app, has viewed available ride requests, and no other request is currently active
Basic Path	1. The Driver selects a ride request from the list of available requests. 2. The system displays the details of the selected request, including the Customer's pickup location, destination, and estimated fare. 3. The Driver confirms acceptance of the ride request. 4. The system assigns the ride to the Driver and removes the request from the list of available requests for other Drivers. 5. The system notifies the Customer that a Driver has accepted the request and provides the Driver's name, vehicle information, and estimated time of arrival. 6. The system provides the Driver with navigation to the Customer's pickup location.
Alternative Paths	In step 3, if the Driver does not confirm the acceptance within a certain time limit, the system returns the request to the list of available requests for other Drivers.

Postcondition	The ride request has been accepted and the Driver is navigating to the Customer's pickup location.
Exception Paths	The Driver may cancel the accepted request before arriving at the pickup location. The Customer will be notified and a new Driver will be assigned.
Other	Once a request has been accepted, the Driver will not receive any other ride requests until the current ride is completed.

3.2.11 Arrive at Pickup

Field	Description
Use Case Name	Arrive at Pickup
Xref	Section 2.2.2 Driver Use Cases
Trigger	The Driver arrives at the Customer's pickup location
Precondition	The Driver has accepted a ride request and has navigated to the Customer's pickup location

Basic Path	1. The Driver arrives at the Customer's pickup location. 2. The Driver selects the arrived option in the KiwiGo driver app. 3. The system notifies the Customer that the Driver has arrived at the pickup location. 4. The system begins a waiting timer for the Customer to arrive at the vehicle. 5. The Customer arrives at the vehicle, and the Driver confirms the pickup by selecting the confirm pickup option in the app. 6. The system records the pickup time and updates the ride status to **in progress** in the database.
Alternative Paths	In step 4, if the Customer does not arrive within 10 minutes of the Driver's arrival, the Driver may cancel the pickup. The system notifies the Customer that the ride has been cancelled and the Driver is now available for other requests.
Postcondition	The Customer has been picked up and the ride is now in progress.
Exception Paths	The Driver may cancel the pickup at any time while waiting for the Customer. The Customer will be notified of the cancellation.

Other	The system monitors Drivers who do not arrive at the pickup location within the estimated time of arrival. Repeated delays may result in a warning or account review.

3.2.12 Complete Ride

Field	Description
Use Case Name	Complete Ride
Xref	Section 2.2.2 Driver Use Cases
Trigger	The Driver arrives at the Customer's destination and drops off the Customer
Precondition	The ride is in progress and the Driver has navigated to the Customer's destination
Basic Path	1. The Driver arrives at the Customer's destination. 2. The Customer exits the vehicle. 3. The Driver selects the complete ride option in the KiwiGo driver app.

	4. The system records the drop off time and calculates the final fare based on the actual distance and duration of the ride. 5. The system processes the payment from the Customer's saved payment method. 6. The system notifies the Customer that the ride is complete and displays the final fare. 7. The system prompts both the Customer and Driver to rate each other. 8. The system updates the ride status to completed in the database. 9. The Driver is now available to receive new ride requests.
Alternative Paths	In step 4, if the final fare differs significantly from the estimated fare, the system displays a detailed breakdown of the charges to the Customer.
Postcondition	The ride has been completed, the payment has been processed, and both the Customer and Driver have been prompted to rate each other.
Exception Paths	If the Driver is unable to complete the ride due to an emergency or vehicle issue, the Driver may cancel the ride in progress and the system will notify the Customer and arrange a new Driver.

Other	The system monitors Drivers who do not reach the destination within the estimated travel time. Repeated delays may result in a warning or account review.

3.2.13 Rate Customer

Field	Description
Use Case Name	Rate Customer
Xref	Section 2.2.2 Driver Use Cases
Trigger	The Driver is prompted to rate the Customer after the ride has been completed
Precondition	The ride has been completed and the Driver has marked the ride as complete in the app
Basic Path	1. The system displays a rating prompt to the Driver after the ride is completed. 2. The Driver selects a rating between 1 and 5 stars for the Customer. 3. The Driver may optionally enter a written comment about the ride experience. 4. The Driver submits the rating.

	5. The system saves the rating and updates the Customer's overall rating in the database. 6. The system confirms to the Driver that the rating has been submitted. 7. The Driver is now available to receive new ride requests.
Alternative Paths	In step 2, if the Driver selects a rating of 1 or 2 stars, the system prompts the Driver to provide additional details about the unsatisfactory experience.
Postcondition	The Customer's rating has been updated in the system and the ride record is marked as fully complete.
Exception Paths	The Driver may skip the rating prompt at any time. The ride will still be marked as complete and the Driver will be available for new requests.
Other	The system will not pair a Driver and Customer together on future rides if either party has previously rated the other as unsatisfactory. Customer ratings are used by the system to monitor behavior and may result in account suspension if ratings are consistently low.

3.3 Logical Structure of the Data

The logical structure of the data to be stored in the KiwiGo database is given below.

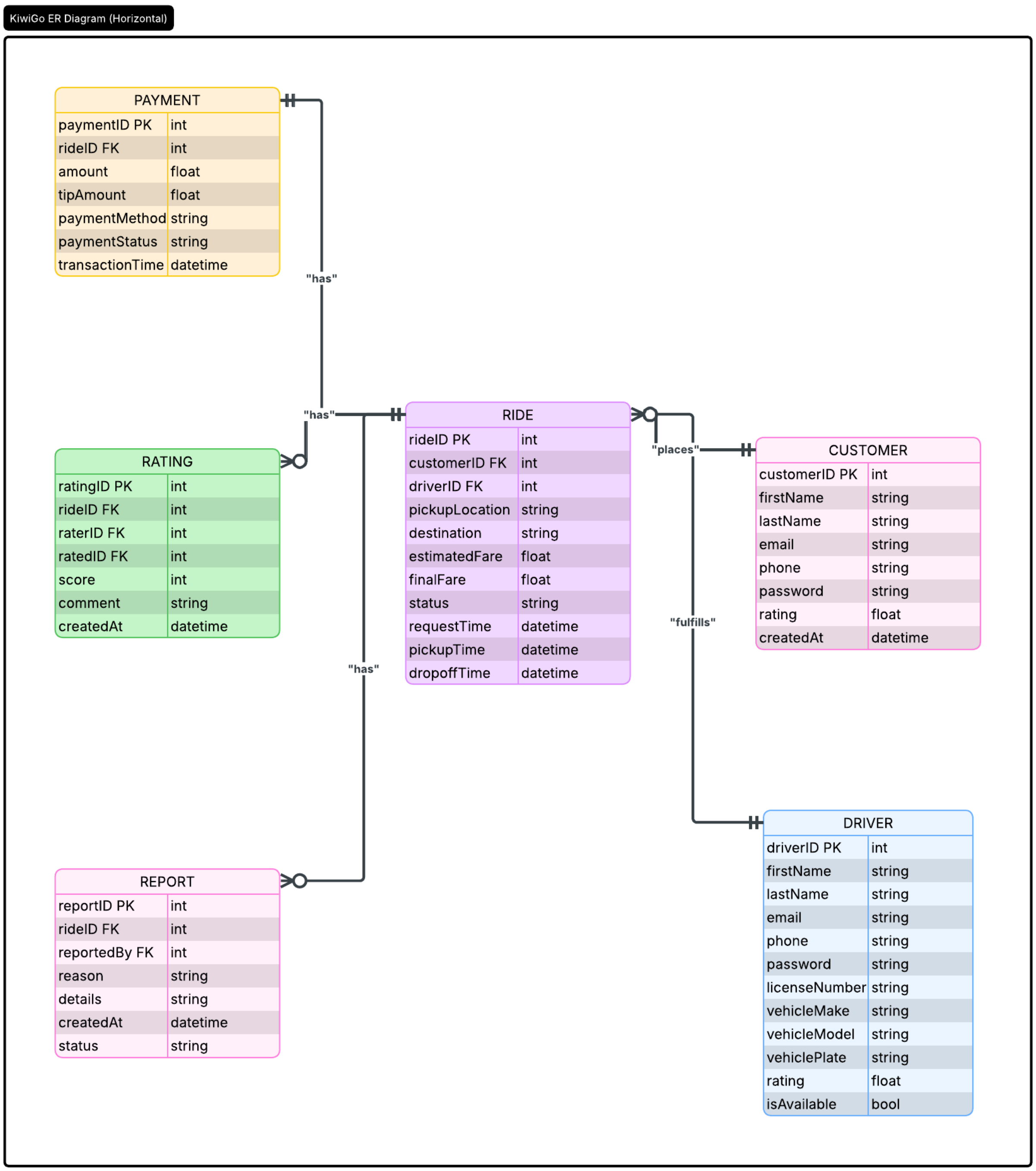

Figure 2 - Logical Structure of the Data

The data element descriptions for each of these data entities are as follows:

Customer Data Entity

Data Item	Type	Description	Comment
customerID	Integer	Unique identifier for each customer	Primary key
firstName	Text	Customer's first name	
lastName	Text	Customer's last name	
email	Text	Customer's email address	Must be unique
phone	Text	Customer's phone number	
password	Text	Customer's encrypted password	
rating	Float	Customer's average rating	Based on Driver ratings
createdAt	DateTime	Date and time account was created	

Driver Data Entity

Data Item	Type	Description	Comment
driverID	Integer	Unique identifier for each driver	Primary key

firstName	Text	Driver's first name	
lastName	Text	Driver's last name	
email	Text	Driver's email address	Must be unique
phone	Text	Driver's phone number	
password	Text	Driver's encrypted password	
licenseNumber	Text	Driver's vehicle license number	
vehicleMake	Text	Make of the Driver's vehicle	
vehicleModel	Text	Model of the Driver's vehicle	
vehiclePlate	Text	License plate of the Driver's vehicle	
rating	Float	Driver's average rating	Based on Customer ratings
isAvailable	Boolean	Whether the Driver is currently available	

Ride Data Entity

Data Item	Type	Description	Comment
rideID	Integer	Unique identifier for each ride	Primary key
customerID	Integer	Reference to the Customer	Foreign key
driverID	Integer	Reference to the Driver	Foreign key
pickupLocation	Text	Customer's pickup location	

destination	Text	Customer's destination	
estimatedFare	Float	Estimated fare before the ride	
finalFare	Float	Final fare after the ride is completed	
status	Text	Current status of the ride	
requestTime	DateTime	Time the ride was requested	
pickupTime	DateTime	Time the Customer was picked up	
dropoffTime	DateTime	Time the Customer was dropped off	

Payment Data Entity

Data Item	Type	Description	Comment
paymentID	Integer	Unique identifier for each payment	Primary key
rideID	Integer	Reference to the associated ride	Foreign key
amount	Float	Total fare amount charged	
tipAmount	Float	Tip amount added by the Customer	Optional
paymentMethod	Text	Payment method used	credit card, debit card

paymentStatus	Text	Status of the payment	pending, completed, failed
transactionTime	DateTime	Time the payment was processed	

Rating Data Entity

Data Item	Type	Description	Comment
ratingID	Integer	Unique identifier for each rating	Primary key
rideID	Integer	Reference to the associated ride	Foreign key
raterID	Integer	ID of the person giving the rating	Foreign key
ratedID	Integer	ID of the person being rated	Foreign key
score	Integer	Rating score between 1 and 5	
comment	Text	Optional written comment	May be null
createdAt	DateTime	Date and time rating was submitted	

Report Data Entity

Data Item	Type	Description	Comment
reportID	Integer	Unique identifier for each report	Primary key
rideID	Integer	Reference to the associated ride	Foreign key
reportedBy	Integer	ID of the person filing the report	Foreign key
reason	Text	Reason for the report	

details	Text	Additional details about the incident	May be null
createdAt	DateTime	Date and time report was submitted	
status	Text	Current status of the report	pending, reviewed, resolved

3.4 Nonfunctional Requirements

These requirements apply to the KiwiGo system as a whole.

Performance: The system should respond to most user requests within 2 seconds during normal use and within 5 seconds during busy periods. The app should support up to 10,000 users at once without major slowdowns. GPS tracking should update the driver's location every 5 seconds, and fare estimates should appear within 3 seconds after entering a destination.

Availability: KiwiGo should have a target uptime of 99.9% each month. System updates should avoid interrupting active rides when possible. If the system fails, service should be restored within 30 minutes.

Security: All data sent between the app and server should be protected with TLS 1.3. Customer and driver passwords should be stored securely. Payments should be handled through a secure third-party payment service, and full card data should not be stored in the KiwiGo database.

Reliability: If a network issue happens during a ride, the system should try to reconnect without losing ride data. Ride, payment, and report records should be backed up daily.

Usability: The app should be easy to use for people with basic smartphone experience, work on both iOS and Android, and provide clear error messages when needed.